

This guide was originally written by Matteo and then translated and revamped by Carlos. You can find the original version in Spanish in Matteo's LinkedIn profile.

We decided to create this for educational purposes. We are not responsible for any misuse that may be given to it.

Legal Notice: It is important to remember that scanning ports without authorization can be illegal. To avoid legal problems, in this guide we will use scanme.nmap.org, a service provided by Nmap for testing and practice purposes.

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Chapter 1:

What is it and what is it used for?

What is Nmap?

Nmap (**Network Mapper**) is an open-source tool used for network scanning and auditing. Its main function is to discover devices, services, and open ports on a network, allowing you to analyze its status and detect possible vulnerabilities. It is widely used in the field of cybersecurity, system administration, and penetration testing.

What is Nmap used for?

These will be some of the uses we will give to Nmap in this guide:

- **Host discovery:** Identifies devices connected to a network.
- **Port scanning:** Detects open ports and services on devices.
- **Operating system detection:** Attempts to identify the operating system and its version.
- **Service recognition:** Determines which services and versions are running on open ports.
- **Security auditing:** Allows detecting insecure configurations and possible vulnerabilities.
- **Automation with scripts (NSE - Nmap Scripting Engine):** Executes scripts to detect vulnerabilities, perform simulated attacks, and much more.

Chapter 2:

Requirements and Installation

To start with this guide, we must prepare our laboratory neatly and cleanly. We will begin by installing the tool. The first thing we need to know is what environment we are going to work in. In my case, it will be from Kali Linux, where Nmap comes pre- installed.

If we want to check if Nmap is installed, we can execute the following command:

```
user@kali:~$ nmap --version
Nmap version 7.94SVN (https://nmap.org)
Platform: x86_64-pc-linux-gnu
Compiled with: liblua-5.4.6 openssl-3.0.13 libssh2-1.11.0 libz-1.3
libpcre2-10.42 libpcap-1.10.4 nmap-libdnet-1.12 ipv6
Compiled without:
Available nsock engines: epoll poll select
```

As we can see, I have version 7.94.

In case you do not have Nmap installed, we can do it as follows:

```
user@kali:~$ sudo apt install nmap -y
```

The **-y** option is used to automatically accept the installation terms, saving us time in the process.

Once installed, we can verify the installation with the command mentioned earlier or by executing the tool with:

```
user@kali:~$ nmap
```

Perfect! We now have our tool installed. Now, how do we use it?

Chapter 3:

How to use Nmap

Well, first of all, we need a victim to test the options we will see later. In this case, the creators of Nmap themselves offer us a victim (themselves), whose URL is: <http://scanme.nmap.org/>

If we had a personal machine for testing, we would use its IP address. In my case, I will use scanme.nmap.org.

Once we have our victim, we must explore the available options until we find the one that interests us, something we will see in more detail later.

For now, we will use the **-sS** option, which, in short, allows us to identify which ports are open quickly and stealthily.

Now that we have the victim and the option to execute, the next step is to combine both elements and execute the tool. The correct order of the parameters is as follows:

- Permissions (in case of not being root): **sudo**
- Tool name: **nmap**
- Option: **-sS**
- IP or URL: scanme.nmap.org

In the terminal, the command would look like this:

```
user@kali:~$ sudo nmap -sS scanme.nmap.org
```

Great, we have everything set up. How about we execute the command and see what it tells us:

```

user@kali:~$ sudo nmap -sS scanme.nmap.org
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-02-11 18:52 -03

Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.23s latency).

Other addresses for scanme.nmap.org (not scanned):
2600:3c01::f03c:91ff:fe18:bb2f

Not shown: 996 closed tcp ports (reset)

PORT      STATE      SERVICE
22/tcp    open      ssh
80/tcp    open      http
9929/tcp  open      nping-echo
31337/tcp open      Elite

Nmap done: 1 IP address (1 host up) scanned in 2.51 seconds

```

Perfect. Before giving you the answers, I would like you to try to analyze it on your own and mentally answer the following questions:

1. What version of Nmap do I or do you have?
2. What IP address does scanme.nmap.org use?
3. How many ports are closed?
4. What ports are open and what are their services?
5. How long did it take to scan?

I hope you can answer them. If not, do not worry; below you will see the answers.

If you managed to answer them, congratulations! If not, do not worry, we will reinforce this later, and if necessary, read it again as many times as needed.

The answers are:

1. What version of Nmap do I have?

The version of Nmap I have is 7.94SVN.

2. What IP address does scanme.nmap.org use?

The IP address of scanme.nmap.org is 45.33.32.156.

3. How many ports are closed?

There are 996 closed ports according to the report.

4. What ports are open and what are their services?

The open ports are:

- 22/tcp (service: ssh)
- 9929/tcp (service: nping-echo)
- 80/tcp (service: http)
- 31337/tcp (service: Elite)

5. How long did it take to scan?

The scan took 2.51 seconds.

Chapter 4:

Essential concepts before Nmap

Well, it is important to understand what happens when launching our scan. Nmap works by sending and analyzing network packets. Depending on the scan type, it will use different protocols and types of packets.

Relax, here is a small list of the communication protocols and network packets:

Communication Protocols

Protocols define how devices communicate on a network. Nmap mainly uses these:

Protocol	Description
TCP (Transmission Control Protocol)	Connection-oriented protocol, meaning it guarantees data delivery in the correct order. It uses the "Three-Way Handshake" (SYN, SYN-ACK, ACK) to establish a connection.
UDP (User Datagram Protocol)	Connectionless protocol, it is faster than TCP, but does not guarantee data delivery nor verify if it has arrived correctly. It is used in services like DNS, VoIP, and streaming.
SCTP (Stream Control Transmission Protocol)	Used to detect open ports in services that employ the SCTP protocol, common in telecommunications, LTE/5G and VoIP. They allow discovering active services, evaluating infrastructure security, and evading TCP/UDP-based detections.

Chapter 5:

Options and their description

Well, the time has come to see the options Nmap offers. We will try to review them all and provide a description of each.

These are the basic Nmap parameters:

Scan Parameter	Description
-sS (SYN Scan)	-sS Also known as a stealth scan, this is Nmap's default scanning option when executed with administrative privileges. It sends SYN packets to initiate a connection without completing it, making the scan less detectable in system logs. Fast and effective.
-sT (TCP Connect Scan)	Performs a full TCP connection with the target host. It is easier to detect in system logs but does not require administrative privileges.
-sU (UDP Scan)	Scans UDP ports. It is generally slower than TCP scanning due to the lack of acknowledgments in UDP communication and the need to wait for ICMP "port unreachable" responses.
-sA (ACK Scan)	Sends TCP packets with the ACK flag set to determine whether a firewall is filtering ports. It does not identify open ports, only filtered or unfiltered states.
-sW (Window Scan)	Similar to the ACK scan, but analyzes the TCP window size to distinguish filtered from unfiltered ports.
-sM (Maimon Scan)	A variation of the FIN scan designed to bypass certain firewall and packet filtering implementations.
-sN (NULL Scan)	Sends TCP packets with no flags set. It can sometimes evade firewall rules and detection systems.
-sF (FIN Scan)	Sends TCP packets with the FIN flag set. It may bypass certain filtering and intrusion detection systems.
-sX (Xmas Scan)	Sends TCP packets with the FIN, PSF, and URG flags enabled. The combination of flags resembles a lit Christmas tree, which gives the scan its name.
-sY (SCTP INIT Scan)	Similar to a SYN scan, but used with the SCTP protocol instead of TCP.
-sZ (SCTP COOKIE ECHO Scan)	An SCTP scanning technique that uses cookies verification to identify open SCTP ports.

We can also find out what version and operating system the victim has:

Parameter	Description
<code>-O</code>	Attempts to detect the target's operating system by analyzing response packets.
<code>--osscan-guess</code>	Forces a more aggressive estimation of the operating system in case of uncertainty.
<code>-sV</code>	Service version scanning. Identifies the specific version of services on open ports.
<code>--version-intensity <level></code>	Adjusts the aggressiveness of the version scan. It ranges from 0 (minimum) to 9 (maximum).

Chapter 6:

Putting the options to the test

Options for the TCP protocol (-sS, -sT, -sA, -sW and -sM)

What is TCP?

TCP (Transmission Control Protocol):

- **Connection-oriented:** Establishes a connection before transmitting data and guarantees ordered and error-free delivery.
- **Reliability:** If a packet is lost or arrives corrupted, TCP retransmits it automatically.
- **Flow and congestion control:** Prevents the network from becoming overloaded with excessive data transmission.
- **Usage:** Ideal for applications that require reliable communication, such as web browsing (HTTP), email services (SMTP), and file transfers (FTP).

Option -sT

Well, now that we have seen the main ones, let's put some options to the test and see their results.

We will go in order. We already tried the -sS option, now let's try the -sT option, which is a bit easier to track, but the advantage is that we do not require root permissions.

```
user@kali:~$ sudo nmap -sT scanme.nmap.org
```

Well, once the command is launched, we must wait to get a response.

```

user@kali:~$ sudo nmap -sT scanme.nmap.org
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-02-12 14:11 -03

Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.23s latency).

Other addresses for scanme.nmap.org (not scanned):
2600:3c01::f03c:91ff:fe18:bb2f

Not shown: 992 closed tcp ports (conn-refused)

PORT      STATE      SERVICE
22/tcp    open       ssh
25/tcp    filtered   smtp
80/tcp    open       http
135/tcp   filtered   msrpc
139/tcp   filtered   netbios-ssn
445/tcp   filtered   microsoft-ds
9929/tcp  open       nping-echo
31337/tcp open       Elite

Nmap done: 1 IP address (1 host up) scanned in 24.47 seconds

```

Well, in my case we managed to get the following result, previously we practiced analyzing the results, so I will skip this part and organize the results we can see.

The scan took 24.47 seconds, had a total of 992 closed ports, 4 open ports, and 4 filtered ports.

Starting with the open ports we have:

Port	State	Service
22/tcp	open	ssh
80/tcp	open	http
9929/tcp	open	nping-echo
31337/tcp	open	Elite

Now let's organize the filtered ports, but before that, what does it mean for the port to be filtered?

Well, it most likely means that the port is guarded by a firewall which blocks or restricts access to these services.

Port	State	Service
25/tcp	filtered	smtp
135/tcp	filtered	msrpc
139/tcp	filtered	netbios-ssn
445/tcp	filtered	microsoft-ds

Option -sA

The time has come to use the **-sA** option.

```
user@kali:~$ sudo nmap -sA scanme.nmap.org
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-02-15 17:51 -03

Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.24s latency).

Other addresses for scanme.nmap.org (not scanned):
2600:3c01::f03c:91ff:fe18:bb2f

Not shown: 1000 filtered tcp ports (no-response)

Nmap done: 1 IP address (1 host up) scanned in 4.82 seconds
```

In this example, Nmap reports:

"Not shown: 1000 filtered tcp ports (no-response)"

This means the target did not respond to the ACK packets, which strongly suggests that a firewall or packet filtering system is silently dropping the packets.

The ACK Scan (-sA) is primarily used to determine whether a firewall is filtering traffic. Unlike SYN scans, ACK scans do not identify open ports. Instead, they help distinguish between filtered and unfiltered states.

Possible ACK scan results include:

- **Filtered** → No response is received, or an ICMP unreachable message is returned. This usually indicates that a firewall is filtering the traffic.
- **Unfiltered** → The target responds with a TCP RST packet, indicating that the port is reachable, although not necessarily open.

In our case, all ports were reported as filtered.

A useful option for investigating the presence of a firewall on specific ports is:

```
user@kali:~$ sudo nmap -sA -p 80,443 scanme.nmap.org
```

In this example, we use the **-p** option to scan ports 80 and 443 and determine whether they are being filtered by a firewall.

```
user@kali:~$ sudo nmap -sA -p 80,443 scanme.nmap.org
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-02-15 17:51 -03

Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.24s latency).

Other addresses for scanme.nmap.org (not scanned):
2600:3c01::f03c:91ff:fe18:bb2f

PORT      STATE      SERVICE
80/tcp    filtered  http
443/tcp   filtered  https

Nmap done: 1 IP address (1 host up) scanned in 4.82 seconds
```

And the result shows that the port is filtered, which lets us understand that we have some type of security system.

Option -sW

The Window Scan (**-sW**) is similar to the ACK Scan (**-sA**), but it analyzes the TCP window size field in RST responses to help distinguish between filtered and unfiltered ports on certain operating systems.

The TCP window size is a value used for flow control. It indicates how much data a system can receive before requiring an acknowledgment (ACK). On some older TCP implementations:

- Window size = 0 → The port is likely closed.
- Window size ≠ 0 → The port may be open.

Because of this behavior, Nmap could sometimes infer port states by examining the returned window size values.

Today, most modern operating systems and firewalls normalize these responses, so Window Scans often produce results very similar to ACK Scans.

Let's perform a basic Window Scan:

```
user@kali:~$ sudo nmap -sW scanme.nmap.org
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-02-15 20:17 -03

Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.28s latency).

Other addresses for scanme.nmap.org (not scanned):
2600:3c01::f03c:91ff:fe18:bb2f

All 1000 scanned ports on scanme.nmap.org (45.33.32.156) are in
ignored states.
Not shown: 1000 filtered tcp ports (no-response)

Nmap done: 1 IP address (1 host up) scanned in 286.76 seconds
```

The result is very similar to the ACK Scan (**-sA**). Nmap reports that all scanned ports are filtered, which suggests that a firewall or filtering device is silently dropping the packets.

We can also scan specific ports or define a port range. In the following example, we scan ports 20 through 100:

```
user@kali:~$ sudo nmap -sW -p 20-100 scanme.nmap.org
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-02-15 20:29 -03

Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.31s latency).

Other addresses for scanme.nmap.org (not scanned):
2600:3c01::f03c:91ff:fe18:bb2f

All 81 scanned ports on scanme.nmap.org (45.33.32.156) are in ignored
states.
Not shown: 81 filtered tcp ports (no-response)

Nmap done: 1 IP address (1 host up) scanned in 24.82 seconds
```

Okay, it seems that within the range all ports are filtered, indicating that Nmap received no response; this, as we saw earlier, indicates the presence of a Firewall. Can we do something else?

Yes, change the option to an **-sA** or change the range. I used the range to show you the different things we can do.

In case you didn't notice when indicating the ports we want to analyze, we can give it a range (example: **10-20**) or list them (example: **10,20,30**).

Option -sM

The Maimon Scan (**-sM**) is a variation of the FIN Scan designed to exploit specific behaviors in older TCP implementations. It sends packets with the FIN and ACK flags set and attempts to identify ports based on how systems respond.

Historically, this scan could sometimes bypass certain firewall or packet filtering configurations. However, its effectiveness on modern operating systems is limited.

Let's perform a Maimon Scan:

```
user@kali:~$ sudo nmap -sM scanme.nmap.org
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-02-15 20:39 -03

Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.23s latency).

Other addresses for scanme.nmap.org (not scanned):
2600:3c01::f03c:91ff:fe18:bb2f

All 1000 scanned ports on scanme.nmap.org (45.33.32.156) are in
ignored states.
Not shown: 1000 open|filtered tcp ports (no-response)

Nmap done: 1 IP address (1 host up) scanned in 236.61 seconds
```

At this point, you may wonder why the result looks similar to previous scans.

The "open|filtered" state means Nmap did not receive enough information to determine whether the ports are open or being filtered. This commonly occurs because:

- A firewall silently drops FIN+ACK packets.
- The operating system ignores these packets instead of responding.
- The target behaves similarly for both open and filtered ports.

As a result, Nmap cannot confidently distinguish between the two states.

Like the Window Scan (-sW), the Maimon Scan (-sM) is less effective against modern systems and often produces ambiguous results. In most practical situations, SYN Scans (-sS) and ACK Scans (-sA) provide more reliable information.

However, Maimon Scans may still be useful when analyzing older systems or unusual TCP implementations.

Option -sN

The NULL Scan (-sN) sends TCP packets with no flags set. Unlike traditional TCP communication, these packets do not initiate a connection and are considered unusual behavior.

According to RFC 793, closed ports should respond with a TCP RST packet, while open ports may ignore the packet and remain silent. Nmap uses this behavior to infer port states.

Because of this:

- RST response → The port is likely closed.
- No response → The port may be open or filtered.

As a result, NULL Scans often return the state:
open | filtered

As we said in the previous option, this means Nmap cannot determine whether the port is actually open or if a firewall is silently dropping the packets.

NULL Scans may sometimes bypass simple packet filters or firewall rules, but modern operating systems and security devices frequently detect or ignore this technique.

Let's perform a NULL Scan:

```
user@kali:~$ sudo nmap -sN scanme.nmap.org
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-02-15 21:08 -03

Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.27s latency).

Other addresses for scanme.nmap.org (not scanned):
2600:3c01::f03c:91ff:fe18:bb2f

All 1000 scanned ports on scanme.nmap.org (45.33.32.156) are in
ignored states.
Not shown: 1000 open|filtered tcp ports (no-response)

Nmap done: 1 IP address (1 host up) scanned in 241.13 seconds
```

The result indicates that Nmap did not receive responses from the target ports. Therefore, it cannot determine whether the ports are open or filtered.

This behavior is common with NULL Scans because many modern systems and firewalls silently ignore packets with no TCP flags set. As a result, the scan often produces ambiguous states such as open | filtered rather than clearly identifying open ports.

Option -sF

Unlike the NULL Scan (-sN), which sends packets with no flags set, the FIN Scan (-sF) sends TCP packets with only the FIN flag enabled.

The FIN flag is normally used to terminate an existing TCP connection. In this case, however, Nmap sends FIN packets without first establishing a connection, relying on how the target system handles these unexpected packets.

Historically, FIN Scans could sometimes bypass simple packet filters focused on SYN traffic. However, modern systems and firewalls often respond similarly to NULL and FIN scans.

For this reason the result would likely be very similar to the previous NULL Scan (-sN), commonly producing states such as:
open | filtered

Option -sX

The Xmas Scan (-sX) is similar to NULL (-sN) and FIN (-sF) scans but sends packets with the FIN, PSH, and URG flags enabled simultaneously.

Like NULL and FIN scans, Xmas Scans rely on how target systems respond to unusual TCP packets.

The Xmas Scan is mainly useful for analyzing legacy TCP behaviors or attempting to bypass simplistic packet filters. On modern systems, it often produces ambiguous open | filtered results similar to NULL and FIN scans.

PORT

Options for the UDP protocol (-sU)

What is UDP?

UDP (User Datagram Protocol):

- **Connectionless:** It does not establish a connection before sending data, making it faster, but it does not guarantee that the data arrives correctly or in the right order.
- **Unreliability:** There are no retries if packets are lost or arrive out of order.
- **Usage:** Used when speed is more important than reliability, such as in live streaming applications (video or audio), online games, or DNS queries.

Now it is our turn for the **-sU** option. For this, we are going to launch it and analyze the result.

Option -sU

```
user@kali:~$ sudo nmap -sU scanme.nmap.org
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-02-13 12:38 -03
Nmap scan report for scanme.nmap.org (45.33.32.156)

Host is up (0.26s latency).

Other addresses for scanme.nmap.org (not scanned):
2600:3c01::f03c:91ff:fe18:bb2f

Not shown: 998 closed udp ports (port-unreach)

PORT      STATE      SERVICE
68/udp    open|filtered dhcpc
123/udp    open       ntp

Nmap done: 1 IP address (1 host up) scanned in 1114.77 seconds
```

Well now let's see what this result presents. As you can see, this result took much longer than usual with 1114.77 seconds (about 18 minutes and 34 seconds), we have a total of 998 closed ports and 2 open ports.

Port	State	Service
68/udp	open filtered	dhcpc
123/udp	open	ntp

In case you are wondering what it means that port 68/udp is both open and filtered at the same time, this indicates that it could not determine with certainty if the port is truly open or if it is being filtered by a Firewall or another security mechanism.

It is something like Schrödinger's cat: it is not really known if it is guarded by any security system. Can we dig deeper to find out for sure? Obviously, but in this first edition we are not going to look at it *(but as additional data, we could find out by analyzing the traffic with Wireshark, to see if there really is a response from our friend port 68 😊)*

Options for the SCTP protocol (-sY and -sZ)

What is SCTP?

SCTP (Stream Control Transmission Protocol):

- **Connection-oriented:** Like TCP, SCTP establishes a connection before transmitting data.
- **Multihoming:** Allows a connection to use multiple IP addresses, improving fault tolerance.
- **Multistreaming:** Allows sending multiple data streams within a single connection, which helps prevent blockages in one stream if another has issues.
- **Usage:** Primarily used in telecommunications applications, such as in mobile telephony networks or in protocols like SIP (Session Initiation Protocol).

The time has come to use Nmap options based on the SCTP protocol.
Let's try the first option **-sY**.

Option -sY

```
user@kali:~$ sudo nmap -sY scanme.nmap.org
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-02-18 12:23 -03
Nmap scan report for scanme.nmap.org (45.33.32.156)

Host is up (0.22s latency).

Other addresses for scanme.nmap.org (not scanned):
2600:3c01::f03c:91ff:fe18:bb2f

All 52 scanned ports on scanme.nmap.org (45.33.32.156) are in ignored
states.

Not shown: 52 closed sctp ports (abort)

Nmap done: 1 IP address (1 host up) scanned in 5.15 seconds
```

Well, looking at the result we already know that the 52 SCTP ports are closed, indicating that the host responded with an abort message, which tells us there are no SCTP services running on the scanned ports.
But let's not just sit back; while we are at it, let's add options.

```
user@kali:~$ sudo nmap -sY -p- scanme.nmap.org
```

Here we add **-p-** to scan all 65,535 possible ports, ensuring that we do not miss anything.

```
user@kali:~$ sudo nmap -sY -p- scanme.nmap.org
Starting Nmap 7.94 ( https://nmap.org ) at 2025-02-28 12:34 UTC
Nmap scan report for scanme.nmap.org (45.33.32.156)
```

```
Host is up (0.12s latency).
```

```
Scanned at 2025-02-28 12:34 UTC with SCTP INIT scan (-sY)
```

PORT	STATE	SERVICE
1/sctp	closed	tcpmux
7/sctp	closed	echo
9/sctp	closed	discard
13/sctp	closed	daytime
17/sctp	closed	qotd
19/sctp	closed	chargen
22/sctp	closed	ssh
23/sctp	closed	telnet
25/sctp	closed	smtp
53/sctp	open	domain
80/sctp	open	http
443/sctp	closed	https
500/sctp	open filtered	isakmp
3306/sctp	closed	mysql
3389/sctp	closed	ms-wbt-server
5060/sctp	closed	sip
8080/sctp	closed	http-proxy

```
Nmap done: 1 IP address (1 host up) scanned in 34.12 seconds
```

Here we received our results, showing a total of 14 closed ports, 2 open ports, and 1 open|filtered port.

However, we can go even further: if the target has a Firewall blocking ping requests, we could add `-Pn` to skip host discovery.

We can also use `--open` to display only open ports and avoid unnecessary results.

```
user@kali:~$ sudo nmap -sY -p- --open -Pn scanme.nmap.org
[sudo] password for user:
Sorry, try again.

[sudo] password for user:

Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-02-28 11:22 -03

Nmap done: 1 IP address (1 host up) scanned in 75.89 seconds
```

The scan did not return any open ports, which indicates that:

- There are no open SCTP ports on scanme.nmap.org.
- The host is still active, since no connectivity errors occurred.
- A Firewall could be blocking SCTP traffic, causing all ports to appear closed or filtered.

The scan using -sY and -p-, --open did not find any open SCTP ports on scanme.nmap.org. This can mean two things:

- The host does not have SCTP services running → No processes are listening on this protocol.
- A Firewall or network policy is blocking SCTP traffic → Some systems completely disable this protocol for security reasons.

Options for further investigation:

If we want to make sure that a Firewall is not blocking the traffic, we can force service detection:

```
user@kali:~$ sudo nmap -sY -p- --open -Pn --reason scanme.nmap.org
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-02-28 11:25 -03
Nmap scan report for scanme.nmap.org (45.33.32.156)

Host is up, received user-set (0.20s latency).

Other addresses for scanme.nmap.org (not scanned):
2600:3c01::f03c:91ff:fe18:bb2f

All 65535 scanned ports on scanme.nmap.org (45.33.32.156) are in
ignored states.

Not shown: 65535 closed sctp ports (abort)

Nmap done: 1 IP address (1 host up) scanned in 1242.36 seconds
```

In this case, Nmap scanned all SCTP ports (-p-) and found that all of them were closed and responded with ABORT, confirming that scanme.nmap.org does not have SCTP services running.

Additionally, since no filtered ports were found, it is unlikely that a Firewall is blocking SCTP traffic.

Alright, we have already gone quite deep into this topic, so it is time to make one final jump to the -sZ option.

Option -sZ

The -sZ scan uses a different approach compared to -sY: instead of sending an SCTP INIT packet, it sends an SCTP COOKIE-ECHO packet to determine whether there are services waiting to complete SCTP authentication.

This method can be useful for detecting services that require verification before fully establishing an SCTP connection.

```
user@kali:~$ sudo nmap -sZ scanme.nmap.org
[sudo] password for user:

Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-02-28 12:08 -03
Nmap scan report for scanme.nmap.org (45.33.32.156)

Host is up (0.20s latency).

Other addresses for scanme.nmap.org (not scanned):
2600:3c01::f03c:91ff:fe18:bb2f

All 52 scanned ports on scanme.nmap.org (45.33.32.156) are in ignored
states.

Not shown: 52 closed sctp ports (abort)

Nmap done: 1 IP address (1 host up) scanned in 2.12 seconds
```

In this case, all ports responded with ABORT, confirming that scanme.nmap.org does not have SCTP services waiting to complete authentication.

This matches the results obtained with -sY, reinforcing that SCTP is not being used on this host.

Option -sV

We could now perform a service scan to identify which services the host is currently running using Nmap.

```
user@kali:~$ sudo nmap -sV scanme.nmap.org
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-03-06 11:19 -03
Nmap scan report for scanme.nmap.org (45.33.32.156)

Host is up (0.21s latency).

Other addresses for scanme.nmap.org (not scanned):
2600:3c01::f03c:91ff:fe18:bb2f

Not shown: 991 closed tcp ports (reset)

PORT      STATE      SERVICE      VERSION
22/tcp    open      ssh          OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13
(Ubuntu Linux; protocol 2.0)
25/tcp    filtered  smtp
80/tcp    open      http         Apache httpd 2.4.7 ((Ubuntu))
135/tcp   filtered  msrpc
139/tcp   filtered  netbios-ssn
445/tcp   filtered  microsoft-ds
9929/tcp  open      nping-echo   Nping echo
31337/tcp open      tcpwrapped
49163/tcp filtered  unknown

Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Nmap done: 1 IP address (1 host up) scanned in 45.59 seconds
```

Now that we know which ports and services are running on the host, we can analyze them further using the different options covered throughout this guide.

With this, we conclude this introductory guide to the basic usage of Nmap. We covered the essential commands needed to begin scanning networks and ports. As you continue learning, you will be able to explore more advanced options and take full advantage of everything this powerful tool has to offer. Always remember to practice ethically and legally. Thank you for reading, and I hope this guide was useful to you.